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Triggering high-energy orbits in rotational nonlinear energy harvesters via time-varying external excitation and internal parameter modulation

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ABSTRACT

The nonlinear dynamics of rotational piezoelectric energy harvesters can enhance power output and broaden the operational bandwidth. However, their practical application is severely constrained by uncontrollable switching between the high-energy orbit (HEO) and low-energy orbit (LEO) induced by nonlinearity and time-varying effects. In addition, centrifugal effects significantly raise the energy threshold required for the harvester to reach the HEO state. To address these challenges, this study proposes an HEO-excitation strategy that combines time-varying external excitation with internal system parameter modulation, enabling synergistic improvements in both power output and operational bandwidth. First, an analytical model of a centrifugal-force-affected rotational harvester is established. A representative nonlinear rotational piezoelectric energy harvester is then designed, and its theoretical model is derived using the method of multiple scales. Next, HEO excitation is investigated through two mechanisms: external rotational frequency modulation and internal electrically induced damping modulation. The corresponding orbit-jump behaviors and performance enhancement effects are validated experimentally. For implementation, rotational frequency modulation is realized through a motor controller, while electrically induced damping modulation is achieved using a bidirectional energy conversion circuit. By inducing and stabilizing the HEO state, the proposed method achieves a maximum 911% increase in output power and a 62.5% expansion of the operational bandwidth. In addition, the harvester operating in the HEO state enables self-powered LoRa-based wireless temperature sensing. Through coordinated parameter modulation, this work presents a new strategy for rotational energy harvesting, providing an effective route for HEO excitation and self-powered sensing.

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1. Introduction

Over the past decade, the Internet of Things (IoT) has flourished and become an important part of the modern information society [1,2]. With the increasing demand for wireless sensing and monitoring, IoT sensor nodes are often deployed in large-scale infrastructures, many of which are remote and have limited access to traditional power grids [3]. Furthermore, their widespread distribution also poses severe challenges for battery replacement and wired power maintenance. Therefore, the development of energy-efficient IoT systems has gained enormous attention for their potential for self-sustainable sensing with smart materials and structures [4–6].

In recent years, vibrational and rotational energy-harvesting technologies have emerged to provide sustainable power sources for IoT systems involving vibration and rotation [6–8]. Vibration energy, which is abundant in machinery such as vehicle engines and bridges, can serve as an important power source for IoT sensors [9]. A significant amount of kinetic energy exists in rotational motion [10,11], which is common in industrial environments such as machinery [12,13], wind turbines [14], and automobile tires [15]. To reduce mechanical vibration noise and prevent structural damage caused by resonance, additional structural damping has become a prevailing practice in rotational machinery [16,17]. In addition to structural damping, recent research efforts have also demonstrated the potential to utilize electrical damping, such as electromagnetic dampers, eddy current dampers, and piezoelectric shunting technologies, to control rotational vibrations. Among them, rotational piezoelectric harvesters have received significant attention due to the possibility of generating electricity with vibration control functions [10]. However, the intrinsic linear dynamics of these devices limit the effective energy-harvesting bandwidth [18,19]. A practical mitigation approach is to introduce nonlinearities to broaden the bandwidth of these devices and thereby extend the operating range of rotational energy harvesting.

Among the various methods for introducing nonlinearity into rotational energy harvesters, the most prevalent types are impact nonlinearity [20,21] and magnetic nonlinearity [22]. For impact nonlinearity, Fang et al. [23] designed a high-performance, compact rotational piezoelectric energy harvester using an internal impact and centrifugal softening mechanism, achieving high energy density and enabling self-powered wireless sensor systems at ultra-low frequencies below 5 Hz. Febbo et al. [24] developed a bistable elastic beam-mass–spring coupled impact piezoelectric vibration energy harvester with tunable frequency peaks, achieving μW-level power output within a rotational frequency range of 0.8–2.2 Hz. Rui et al. [25] proposed a triple-beam multi-impact piezoelectric rotational energy harvesting structure, which was experimentally demonstrated to generate an output power of 52.1 μW at 4.2 Hz with a 1.9 Hz bandwidth above 2 V. For magnetic nonlinear harvesters, Dong et al. [26] proposed a magnetostrictive bistable rotational harvester that broadens the operational bandwidth and highlights the critical role of the magnetic gap in bistable performance. Mei et al. [27] designed a tristable harvester via magnetic nonlinearity, experimentally validating its effectiveness across 240–440 rpm. Wang et al. [28] proposed a flexible bistable harvester with time-varying potential wells under centrifugal force, enabling enhanced operational bandwidth for broadband applications. Compared to impact nonlinearities, magnetic nonlinearities offer contactless interaction, reducing wear and noise, and providing tunable force–distance characteristics for more precise control [29].

However, the introduction of nonlinearities can give rise to multi-solution behaviors: multiple steady-state responses coexist at the same excitation frequency, often with significantly different amplitudes [30,31]. Crucially, HEOs (large-amplitude limit-cycle oscillations) exhibit electromechanical conversion efficiencies 3–7 times those of LEOs [32]. Because this high-energy orbit cannot be achieved spontaneously, controlled transitions from low- to high-energy orbits remain a fundamental challenge in the optimization of nonlinear harvesters [33]. To achieve high-energy orbits, considerable effort has been devoted to this problem over the past decade. For instance, Wang and Liao [30] proposed load perturbation to induce orbital transitions, while Huguet et al. [34] demonstrated buckling-level rapid modulation for achieving orbit jumps. Saint-Martin et al. [35] introduced an optimization framework for bistable harvesters that combines potential well tuning with orbital transition mechanisms and achieves a threefold enhancement in performance. Yan et al. [36] employed the synchronized switch-on-stiffness (SSOS) control [37] for HEO excitation. Similarly, Zhao et al. [38] utilized the bidirectional energy conversion circuit (BECC) to actively inject energy into piezoelectric components, enabling controlled HEO activation. Despite the above orbit-jump solutions, few alternatives have been proposed for rotational piezoelectric energy harvesters. Yan et al. [39] proposed a passive solution for HEO excitation by utilizing a bistable, dual-degree-of-freedom piezoelectric energy harvester design featuring inherent phase differences in rotational systems. While this method successfully induces orbital transitions during system start-up, it fails to actively trigger HEOs once the system has been stabilized in a low-energy state. The challenges of triggering HEOs in rotational energy harvesters stem from two aspects. First, rotational harvesters are subject to gravitational excitation during rotation, which leads to large vibration amplitudes and a high energy threshold required to trigger HEOs. Second, in rotational regimes, frequency-dependent centrifugal forces induce time-varying stiffness and further dynamic uncertainties that make the excitation of HEOs particularly challenging.

Motivated by the above limitations, this work does not simply transfer an existing orbit-jump or BECC-based method to a new device. Instead, it develops a dual-modulation HEO-triggering framework for centrifugal-force-affected rotational nonlinear piezoelectric energy harvesters. Existing orbit-jump approaches, including load perturbation, potential-well modulation, stiffness modulation, and BECC-based energy injection [8,30,35–38], have mainly been developed for conventional nonlinear vibration harvesters. In contrast, only limited studies have addressed HEO triggering in rotational energy harvesters, and the reported rotational solution mainly relies on passive start-up transients [39]. For rotational harvesters, HEO capture is further complicated by gravitational excitation and frequency-dependent centrifugal stiffness, which modify the response branches and attraction basins. To overcome these challenges, two complementary modulation routes are proposed in this work. The external route uses prescribed rotational-frequency variation to guide the system trajectory into the basin of attraction of the HEO, and the corresponding critical

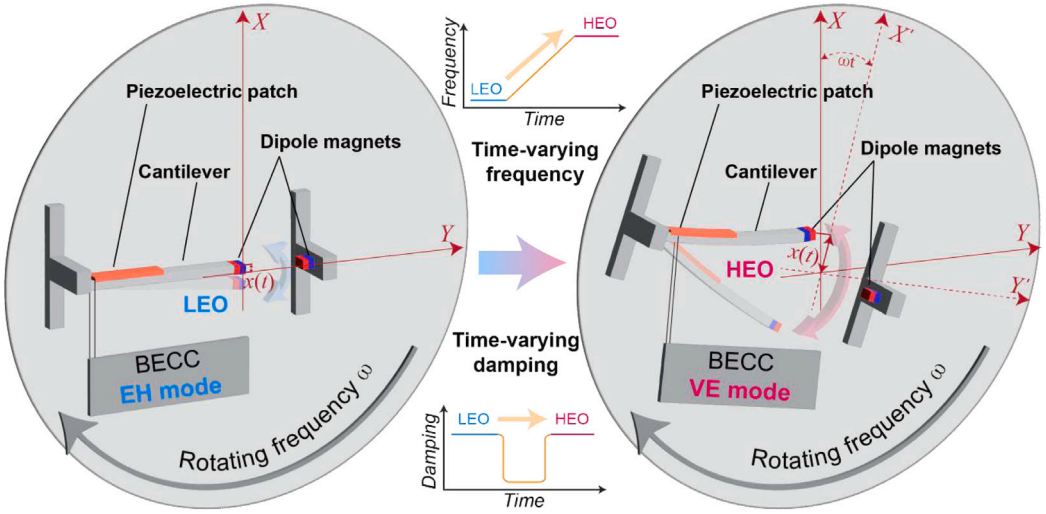


Fig. 1. Schematic of the rotational nonlinear piezoelectric energy harvester mounted on a rotating disk. The transition from the LEO to the HEO can be triggered by two complementary strategies: external time-varying rotational frequency excitation and internal time-varying electrically induced damping modulation.

sweep-rate boundary is identified as a quantitative criterion for successful HEO capture. The internal route uses the BECC not as a newly proposed circuit topology but as a self-powered actuator for temporarily reducing the electrically induced damping, thereby enabling orbit transition even when the rotational speed is fixed. Therefore, the main contribution of this work lies in establishing a dual-modulation strategy for controlled HEO triggering of rotational nonlinear energy harvesters, and the critical sweep-rate analysis provides a practical design criterion for the external modulation route.

Based on this framework, an analytical model incorporating centrifugal stiffness, magnetic nonlinearity, and electrically induced damping is established for the rotational nonlinear piezoelectric energy harvester. The proposed HEO-triggering mechanisms are then investigated through two routes: external time-varying rotational-frequency modulation and internal time-varying electrically induced damping modulation. The corresponding orbit-jump behaviors, critical sweep-rate boundary, and performance enhancement effects are validated experimentally. Finally, a self-powered LoRa wireless sensing system is demonstrated to verify the practical power-supply capability of the harvester operating in the HEO state.

2. Theoretical analysis

A nonlinear rotational piezoelectric energy harvester is considered, as shown in Fig. 1. The system comprises a rotating disk, a piezoelectric cantilever beam, and permanent magnets attached to the tip of the cantilever and the rotating disk. Nonlinearity is introduced by adjusting the distance between the magnets. To establish the governing equations for the centrifugal-softening energy harvester, this study employs the first bending mode of an Euler–Bernoulli beam with a partially covered piezoelectric patch. The lumped model of the clamped piezoelectric beam under a centrifugal load can be represented as:

$$\begin{cases} M\ddot{x} + C\dot{x} + \left(K - K_1 + K_c\omega^2\right)x + K_3x^3 + \alpha_e v_p = \kappa g\cos(\omega t) \\ \alpha_e\dot{x} - C_p\dot{v}_p - i_p = 0 \end{cases} \tag{1}$$

where M, C, and K represent the equivalent mass, damping, and stiffness of the cantilever beam, while K_1 and K_3 denote the coefficients of the nonlinear stiffness caused by the magnetic reaction force. K_c is negative for centrifugal-softening harvesters and positive for centrifugal-stiffening harvesters. α_e is the force-voltage factor for the electromechanical coupling. κ is the modal coefficient of gravitational forcing. g is the gravitational acceleration constant (9.8 m/s^2). C_p is the clamped capacitance of the piezoelectric patch. The displacement of the equivalent mass is denoted by x. As shown in Fig. 1, the X-Y coordinate system is fixed to the rotating disk and lies in the disk plane, in which the displacement x is measured. v_p denotes the voltage across the piezoelectric element. i_p represents the current flowing through the interface circuit. The design in this paper adopts the rotational softening mechanism and focuses on its dynamic analysis; the detailed derivation process is provided in Fang et al. [40]. To quantify the electromechanical dynamics under the electrically induced damping and stiffness through the BECC, Eq. (1) is reformulated as follows:

$$M\ddot{x} + (C + C_e)\dot{x} + (K + K_e - K_1 + K_c\omega^2)x + K_3x^3 = \kappa g\cos(\omega t) \tag{2}$$

where C_e and K_e are the electrically induced damping and stiffness. Because K_e is fixed, this paper focuses on the electrically tunable damping C_e. The corresponding impedance expressions can be found in Zhao et al. [38,41]. To facilitate parametric analysis, Eq. (2)

is rewritten in a dimensionless form:

$$\ddot{x} + c\dot{x} + (\omega_0^2 + k_c\omega^2)x + k_3x^3 = b\cos\omega t \tag{3}$$

The normalization is performed with respect to the system mass M, leading to the following definitions: $\omega_0^2 = (K + K_e - K_1)/M$, $k_3 = K_3/M$, $k_c = K_c/M$, $c = (C + C_e)/M$, $b = \kappa g/M$. Note that ω_0 is the natural frequency of the underlying zero-rotation linear system, k_3 is the normalized cubic stiffness coefficient, k_c is the stiffness contribution from the centrifugal force, c is the normalized total damping (including electrical damping), and b is the normalized amplitude of the gravitational forcing. ω denotes the rotational angular frequency. The analytical solution of Eq. (3) can be approximated as:

$$x(t) = a(t)\cos[\omega t + \phi(t)] + \mathrm{O}(\varepsilon) \tag{4}$$

The amplitude $a(t)$ and phase $\phi(t)$ are slowly varying real-valued functions. This solution can be constructed in terms of an asymptotic series by the method of multiple scales (MST). Two independent time scales $T_0 = t$ (fast) and $T_1 = \varepsilon t$ (slow) are introduced, and the solution can be expanded as:

$$x = x_0\left(T_0, T_1\right) + \varepsilon x_1\left(T_0, T_1\right) + \cdots \tag{5}$$

Since the deviation between the excitation frequency and the intrinsic frequency varies with time, it is expressed as follows:

$$\omega_0^2 + k_c\omega^2 = \omega^2 + \varepsilon\sigma \tag{6}$$

Substituting Eqs. (5) and (6) into Eq. (3) allows us to collect the coefficients of different orders of ε:

$$\begin{cases} \varepsilon^0 : \frac{\partial^2 x_0}{\partial T_0^2} + \omega^2 x_0 = 0 \\ \varepsilon^1 : \frac{\partial^2 x_1}{\partial T_0^2} + \omega^2 x_1 = \bar{b}\cos\left(\omega T_0\right) - \sigma x_0 - 2\frac{\partial^2 x_0}{\partial T_0 \partial T_1} - \bar{c}\frac{\partial x_0}{\partial T_0} - \bar{k}_3 x_0^3 \end{cases} \tag{7}$$

where $\bar{b} = b/\varepsilon$, $\bar{c} = c/\varepsilon$, and $\bar{k}_3 = k_3/\varepsilon$. The solution of the $O(1)$ equation in Eq. (7) is $x_0 = A(T_1)e^{j\omega T_0} + \bar{A}(T_1)e^{-j\omega T_0}$. In this equation, A and $\bar{A}$ are a pair of complex conjugates. Substituting x_0 into the $O(\varepsilon)$ equation in Eq. (7) gives:

$$\frac{\partial^2 x_1}{\partial T_0^2} + \omega^2 x_1 = \text{c.c.} + ST_1 e^{j\omega T_0} - \bar{k}_3 A^3 e^{3j\omega T_0} \tag{8}$$

where c.c. and ST_1 represent the complex conjugate and the secular term, respectively. Setting the coefficient of the secular term to zero and separating the real and imaginary parts yields:

$$\begin{cases} \frac{\mathrm{d}a}{\mathrm{d}t} = -\frac{ac}{2} - \frac{b}{2\omega}\sin\phi, \\ \frac{\mathrm{d}\phi}{\mathrm{d}t} = \frac{\omega_0^2 + (k_c - 1)\omega^2}{2\omega} + \frac{3k_3a^2}{8\omega} - \frac{b}{2\omega a}\cos\phi \end{cases} \tag{9}$$

The slow-flow equations are obtained for the amplitude and phase. Setting $\mathrm{d}a/\mathrm{d}t = 0$ and $\mathrm{d}\phi/\mathrm{d}t = 0$ gives:

$$\begin{cases} \sin\phi = -\frac{ac}{b}\omega \\ \cos\phi = \frac{a\left[\omega_0^2 + (k_c - 1)\omega^2\right]}{b} + \frac{3k_3a^3}{4b} \end{cases} \tag{10}$$

Since $\sin^2\phi + \cos^2\phi = 1$, Eq. (10) yields:

$$\frac{a}{b} = \frac{1}{\sqrt{\left[\left(1 - k_c\right)\omega^2 - \omega_0^2 - \frac{3}{4}k_3a^2\right]^2 + c^2\omega^2}} \tag{11}$$

According to Eq. (11), the analytical solutions for both forward and backward frequency sweeps, the unstable branch, and the two critical jump frequencies can be obtained, as shown in Fig. 2. On this basis, two orbit-jump strategies are further proposed to realize controlled excitation of the HEO. As shown in Fig. 1, the first strategy relies on external time-varying rotational frequency modulation, in which the external excitation trajectory is properly designed to drive the system into the basin of attraction of the HEO. The second strategy is based on internal time-varying electrically induced damping modulation, where the effective damping is temporarily reduced through electrical regulation so that the system response can grow and finally settle on the HEO branch. These two strategies provide complementary routes for inducing orbit transition under rotational operation, and together constitute the overall orbit-jump framework developed in this work.

3. Analysis of time-varying external excitation

The acceleration dynamics of rotating equipment under frequency-sweep excitation play a crucial role in nonlinear rotational energy harvesting systems. Throughout the transition from an initial rotational frequency to the target rotational speed, the transient response characteristics fundamentally govern the ability of the system to stabilize in the HEO, which is the optimal state for maximizing energy conversion efficiency.

Two critical aspects emerge from this dynamic process: The first aspect is the sweep-rate dependence. The final steady-state configuration, whether a high-energy or low-energy orbit, is strongly dependent on the frequency sweep rate. Successful stabilization

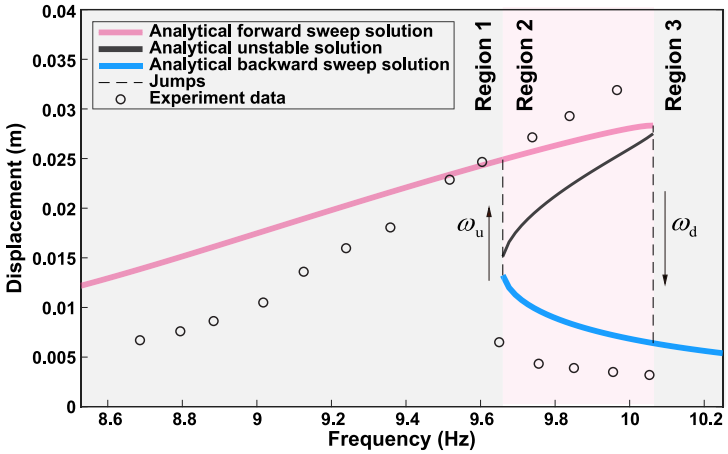


Fig. 2. Theoretical and experimental results of the system frequency response.

in the HEO confirms effective excitation of nonlinear characteristics through optimized sweep rates, whereas convergence to the LEO indicates insufficient modulation to trigger the orbital transition. The second aspect is the critical sweep-rate threshold. There exists a critical frequency sweep rate at the bifurcation point between energy states. This threshold quantifies the maximum allowable frequency-variation rate for successful HEO capture and serves as a key indicator of the system's transient orbit accessibility. Experimental determination of this threshold enables the formulation of optimal acceleration control strategies, which is especially vital for variable-speed energy harvesters (e.g., wind- or hydro-driven systems) where the rotational frequency inherently fluctuates with environmental conditions. This fundamental understanding lays the theoretical foundation for designing controlled acceleration protocols that enable robust transitions to the HEO during rotational-speed variation, thus enhancing the achievable power output in practical energy harvesting applications.

Frequency sweeping of the rotational speed is an effective way to capture the response evolution of the system under time-varying excitation. Nevertheless, in nonlinear rotational systems, the sweep rate is not merely a testing parameter but a key factor that can alter the response path and the accessibility of the HEO. Accordingly, this section investigates how different sweep rates influence the excitation of the HEO.

3.1. Time-varying frequency scheme

The time-varying rotational frequency is expressed as

$$f(t) = f_i + \beta t \tag{12}$$

where f_i is the initial rotational frequency, β denotes the sweep rate in Hz/s, and t is the time in s.

According to Fig. 2, the system frequency response can be classified into three regions. Region 1 corresponds to the low-frequency single-solution range, $f(t) \le \omega_u/(2\pi)$. Region 2 corresponds to the multi-solution range, $\omega_u/(2\pi) < f(t) < \omega_d/(2\pi)$, where two stable solutions and one unstable solution coexist. Region 3 corresponds to the high-frequency single-solution range, $f(t) \ge \omega_d/(2\pi)$.

Because the HEO exists only in Region 2, the terminal frequency of the sweep must fall within this region. In contrast, the initial frequency is chosen in Region 1, since starting from Region 2 or Region 3 causes the system to remain on the LEO branch, making the transition to the HEO unattainable regardless of the sweep rate. Therefore, this study only considers the sweep case from Region 1 to Region 2, denoted by R_{12}, in order to examine the effect of sweep rate on the final response.

The sweeping procedure is specified as follows:

1. The system is first operated at $f_i = 8.5$ Hz until the steady state is reached.
2. The rotational frequency is then increased to f_e with a given sweep rate.
3. The frequency is then kept constant at f_e.

3.2. Modulation results

As shown in Fig. 3, the time-domain response characteristics during a linear frequency sweep from 8.5 Hz to 10 Hz at a sweep rate of 0.5 Hz/s are illustrated through three subplots. Fig. 3(a) presents the frequency-time relationship, visually confirming the linearity of the sweep process. Fig. 3(b) compares the displacement response curves obtained via the method of multiple scales (MST) and the Runge–Kutta method (RKM). The remarkable consistency between these two methodologies validates the accuracy and reliability of the proposed MST in analyzing the dynamic behavior of nonlinear rotational piezoelectric energy harvesting systems. Notably, the sweep initiation and termination points marked in Fig. 3(b) are temporally aligned with the breakpoints in the first plot. Fig. 3(c)

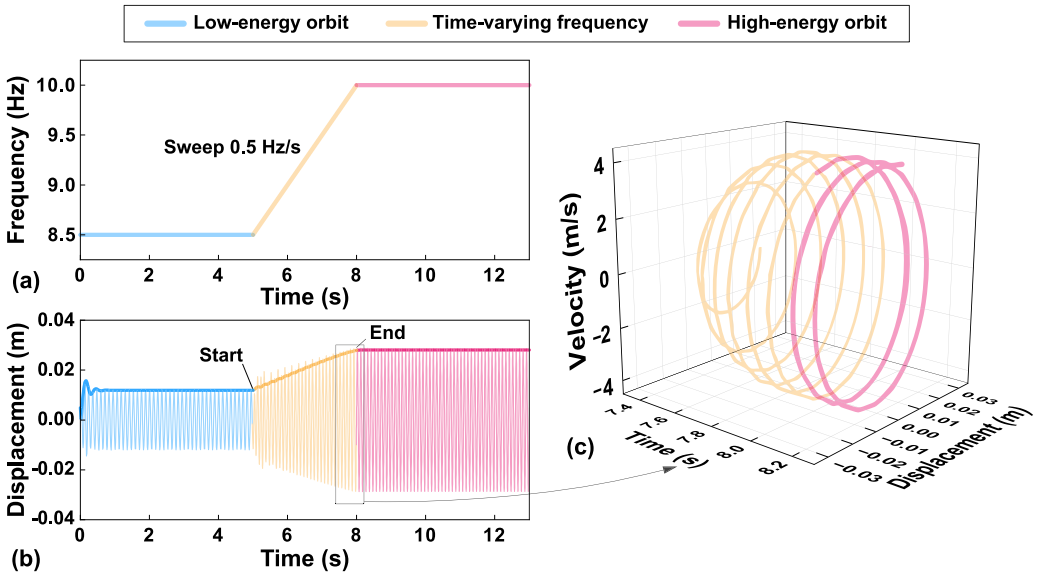


Fig. 3. MST and RKM (envelope curve) simulation results of the displacement response of the system at $\beta = 0.5$ Hz/s: (a) profile of time-varying frequency excitation; (b) time evolution of the displacement response; (c) magnified view of the response profile in (b).

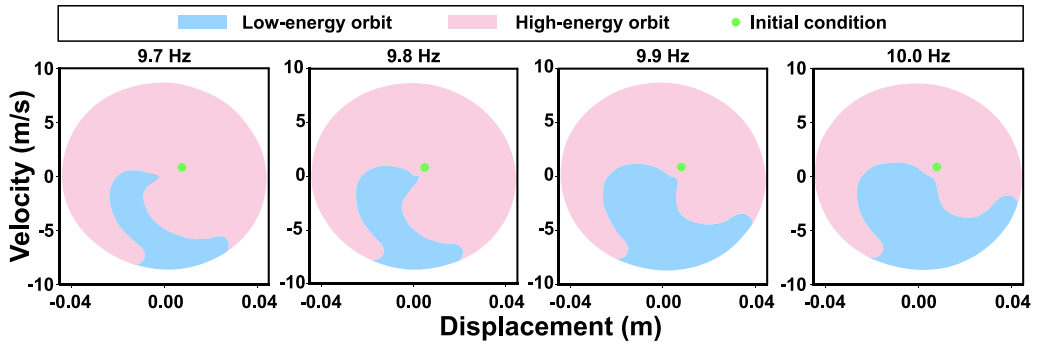


Fig. 4. Evolution of the system's basin of attraction during frequency sweeping from 9.7 Hz to 10.0 Hz at a sweep rate of 0.5 Hz/s in simulation.

shows a three-dimensional magnified view of the black rectangular region in Fig. 3(b), with velocity incorporated as an additional axis to reveal the evolutionary trajectories of both displacement and velocity responses during the sweep process. This visualization offers an intuitive perspective for investigating the system's transient dynamics.

To illustrate the response evolution under time-varying rotational excitation, a representative case with a sweep rate of 0.5 Hz/s is considered. Starting from the steady state at 8.5 Hz, the system response grows continuously during the forward sweep toward 10 Hz, following the high-energy branch of the frequency response. After the sweep terminates, the system remains locked in the HEO at 10 Hz, exhibiting large-amplitude steady-state oscillations with a displacement amplitude of approximately 27 mm. This result indicates that the selected sweep rate is sufficient to drive the system into the HEO and sustain high-energy operation.

The validity of the MST method was further checked using the frequency domain harmonic coefficients of the numerical results. The cubic nonlinear term produces a fundamental resonance shift and higher-harmonic generation. The fundamental component is retained in the slow-flow equations as the nonlinear stiffness correction, whereas the third-harmonic or higher-harmonic components remain bounded non-resonant corrections. We applied a phase-synchronized sliding-window frequency domain projection, retaining harmonic orders up to the seventh harmonic order to compare experimental and numerical displacement records at $\beta = 0.5$ Hz/s. Across all windows, the experimental results yielded the maximum third harmonic generation of 1.52% and a maximum total harmonic distortion of 8.36%, whereas the numerical results of the MST method provided 3.96% and 3.96%, respectively. These conservative extrema confirm that the first harmonic remains dominant while the centrifugal stiffness and excitation frequency vary simultaneously, thereby validating the first harmonic approximation used by the MST of this work.

To further explain why the system can remain in the HEO after the sweep, numerical basin-of-attraction analyses are performed, as shown in Fig. 4. Using the system state near 9.6 Hz as the reference initial condition, the basins are evaluated at 9.7 Hz, 9.8 Hz, 9.9 Hz, and 10.0 Hz. The results show that the basin associated with the HEO gradually shrinks as the frequency increases, implying

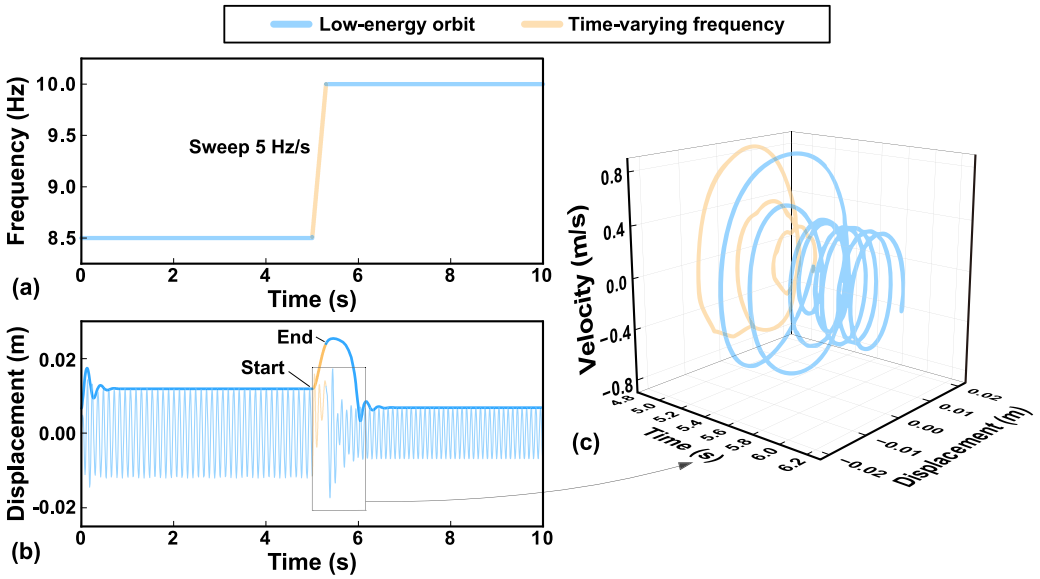


Fig. 5. MST and RKM (envelope curve) simulation results of the displacement response of the system at $\beta = 5$ Hz/s: (a) profile of time-varying frequency excitation; (b) time evolution of the displacement response; (c) magnified view of the response profile in (b).

that the stability region of the HEO becomes progressively smaller. Nevertheless, the actual system trajectory remains inside the high-energy basin throughout the sweep process, which explains why the response can finally settle on the HEO at 10 Hz. This observation is also consistent with the 3D phase evolution shown in Fig. 3, where the trajectory continuously develops toward the high-energy state. Therefore, although the accessible basin of the HEO decreases with increasing frequency, a sweep rate of 0.5 Hz/s still enables the system to successfully enter and maintain the HEO.

At a sweep rate of 0.5 Hz/s, the system achieves sufficient stabilization time at each frequency point, enabling successful HEO capture. This phenomenon can be explained from a dynamical-systems perspective: the lower sweep rate provides sufficient time for the response to relax toward and approximately track the moving HEO branch. To quantitatively investigate the relationship between sweep rates and orbital states (HEOs vs. LEOs), this study extends the analysis by increasing the sweep rate using both MST and RKM. When the sweep rate is increased to 5 Hz/s, rapid frequency variations drastically reduce the stabilization time at each frequency point, as illustrated in Fig. 5. This insufficient stabilization time likely leads to eventual convergence to the LEO. Fig. 5 maintains identical content and layout to Fig. 3, with the sole independent variable being the increased sweep rate from 0.5 Hz/s to 5 Hz/s. Fig. 6 presents the basin-of-attraction evolution under the 5 Hz/s sweep rate, serving as a comparative counterpart to Fig. 4. Comparative analysis reveals that at 9.7 Hz, 9.8 Hz, and 9.9 Hz, despite the pronounced contraction of the high-energy basin area with increasing frequency, the system remains confined within the high-energy basin. However, upon reaching 10.0 Hz, the system leaves the HEO and ultimately settles into the LEO. This critical frequency threshold highlights the dynamic instability induced by excessive sweep rates, where the trajectory fails to remain inside the shrinking basin of attraction of the HEO, resulting in orbital degradation.

The critical sweep rate can be further interpreted from the viewpoint of dynamic bifurcation. For each fixed rotational frequency, the averaged slow-flow equations define stable and unstable fixed points corresponding to the HEO, LEO, and the intermediate saddle-type response. During the frequency-sweep process, these fixed points and their basins of attraction vary with the instantaneous rotational frequency. When the sweep rate is sufficiently low, the system trajectory can approximately follow the moving stable high-energy branch and remains inside the basin of attraction of the HEO. However, when the sweep rate increases, the response may not have enough time to relax toward the moving high-energy attractor before the attraction basin changes.

The local relaxation rate of the amplitude-phase dynamics can be characterized by the eigenvalues of the Jacobian matrix obtained by linearizing the slow-flow equations around the instantaneous fixed points. Near the saddle–node-type transition, the dominant eigenvalue approaches zero, indicating critical slowing down and a significant increase in the local relaxation time. Therefore, successful HEO capture requires that the imposed sweep timescale be longer than the intrinsic relaxation timescale of the nonlinear response. If the sweep timescale becomes shorter than this relaxation time, the trajectory lags behind the moving HEO branch and may cross the basin boundary associated with the unstable saddle structure. In this case, the system finally converges to the LEO after the sweep is completed. Hence, the critical sweep rate is not merely an empirical parameter, but represents the maximum frequency-variation rate allowed by the dynamic relaxation capability of the nonlinear system.

4. Time-varying internal damping analysis

In contrast to the external time-varying frequency excitation discussed above, this section focuses on the second orbit-jump strategy, namely internal time-varying electrically induced damping, and investigates its effect on the accessibility of the HEO. For

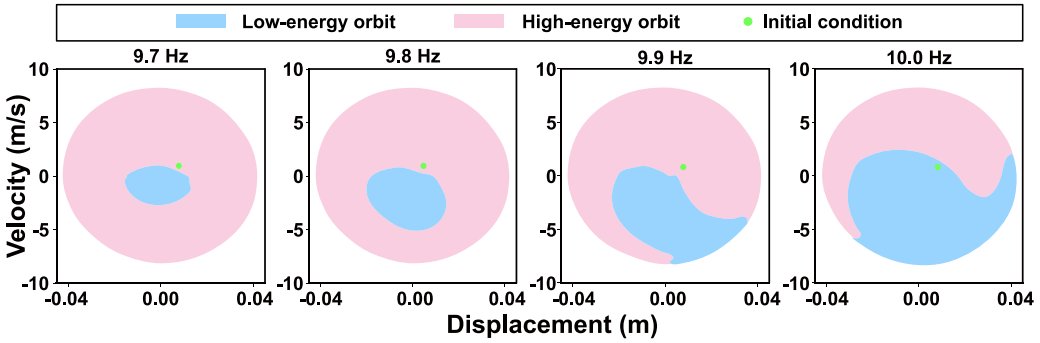


Fig. 6. Evolution of the system's basin of attraction during frequency sweeping from 9.7 Hz to 10.0 Hz at a sweep rate of 5 Hz/s in simulation.

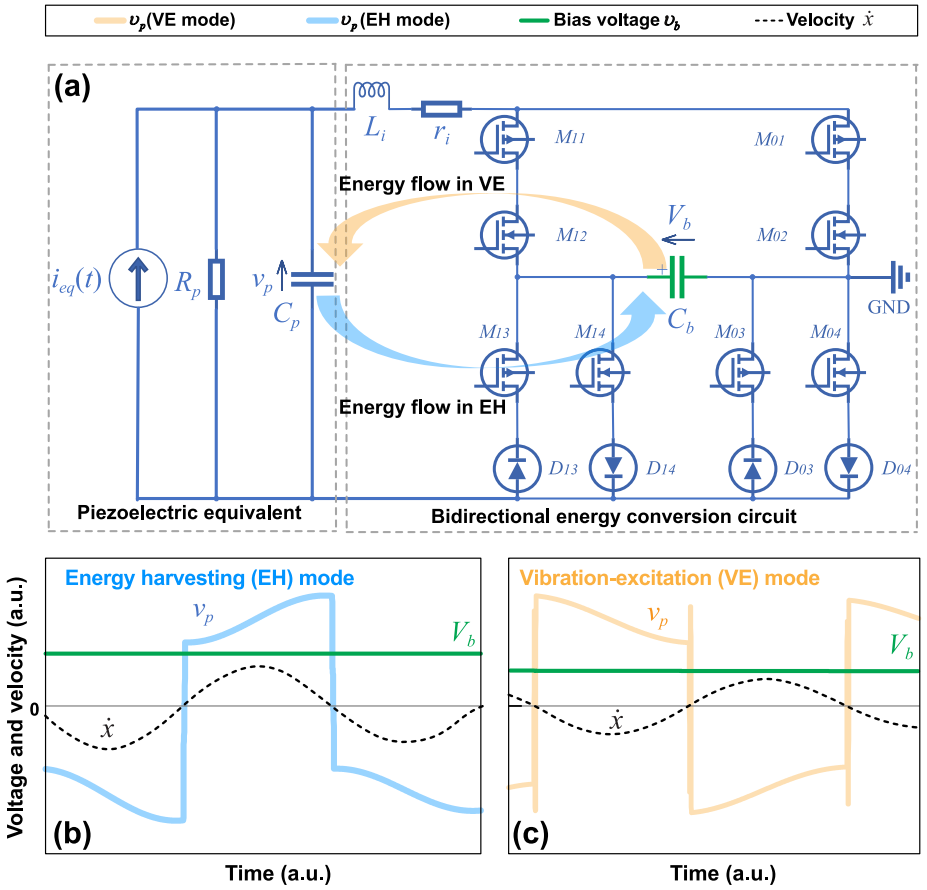


Fig. 7. Schematic of the bidirectional energy conversion circuit (BECC) and its operating modes: (a) circuit topology; (b) piezoelectric voltage and switching waveforms in the energy-harvesting (EH) mode; (c) piezoelectric voltage and switching waveforms in the vibration-excitation (VE) mode.

rotational energy harvesting systems, this internally regulated damping modulation is analyzed through phase portraits to reveal its role in inducing rotational orbital jumps. In this work, the modulation is realized by incorporating the bidirectional energy conversion circuit (BECC), proposed in Zhao et al. [38,42], into the piezoelectric harvester. As shown in Fig. 7, the BECC can be configured in energy-harvesting (EH) mode and vibration-excitation (VE) mode. In particular, the VE mode introduces negative electrically induced damping, which reduces the effective damping of the system and promotes response growth. Therefore, the temporal regulation of internal electrically induced damping through the BECC provides a practical route for achieving controlled transitions from the LEO to the HEO.

4.1. Time-varying damping scheme

The combined topology of the piezoelectric element and the bidirectional energy conversion circuit (BECC) is shown in Fig. 7(a). The left-hand side represents the equivalent lumped-parameter model of the piezoelectric element near its fundamental vibration mode, and the right-hand side shows the BECC interface circuit. Since the detailed circuit operation and switching process have been presented in our previous work [42], only the key results are summarized here.

The current through the piezoelectric branch can be written as:

$$i_{eq}(t) = \alpha_e \dot{x}(t), \tag{13}$$

where α_e is the electromechanical coupling coefficient and $\dot{x}(t)$ is the vibration velocity of the rotational energy harvester. Under the assumption of weak nonlinearity and a first-harmonic dominant response, the piezoelectric voltage v_p can still be approximated as a first-order resonant periodic signal. Based on the fundamental harmonic relationship between the piezoelectric voltage and current, the equivalent impedance of the combined piezoelectric clamped capacitance C_p and BECC can be expressed as:

$$Z_e(j\omega) = \frac{V_{p,f}(j\omega)}{I_{eq}(j\omega)} = \frac{1}{\omega C_p}\left[\frac{4}{\pi}\left(1-\tilde{V}_M\right) - j\right], \tag{14}$$

where $V_{p,f}$ is the fundamental harmonic of the piezoelectric voltage v_p, and $\tilde{V}_M = V_M/V_{oc}$ is the normalized final voltage after the synchronized bias-flip process. Here, V_M denotes the final value of the piezoelectric voltage immediately after M bias-flip actions at each synchronized switching instant [42], and V_{oc} is the open-circuit voltage, given by:

$$V_{oc} = \frac{1}{2C_p}\int_0^{\pi/\omega} i_{eq}(t)\,\mathrm{d}t. \tag{15}$$

According to the electromechanical analogy, the equivalent impedance Z_e introduces electrically induced damping and stiffness into the mechanical domain, which can be written as:

$$C_e = \alpha_e^2 \,\mathrm{Re}\{Z_e\}, \qquad K_e = -\omega\alpha_e^2 \,\mathrm{Im}\{Z_e\}. \tag{16}$$

Substituting Eq. (14) into the above expressions yields:

$$C_e = \frac{4\alpha_e^2}{\omega\pi C_p}\left(1-\tilde{V}_M\right), \qquad K_e = \frac{\alpha_e^2}{C_p}. \tag{17}$$

It can be seen that K_e remains constant, whereas C_e depends on the normalized final voltage $\tilde{V}_M$, and therefore is tunable through the operating condition of the BECC.

In the VE mode, as shown in Fig. 7(c), part of the electrical energy harvested by the piezoelectric element and stored in the bias capacitor C_b is fed back to the piezoelectric structure during synchronized switching. As a result, $\tilde{V}_M > 1$, which leads to $C_e < 0$. This means that the BECC introduces a negative electrically induced damping effect that effectively reduces the total damping of the system. More importantly, since this negative damping is supported by the energy harvested from the system vibration itself, the corresponding damping modulation does not rely on an external power supply but instead forms a self-powered internal parameter regulation mechanism. For the $M = 3$ bias-flip case considered in this work, namely the S3BF VE mode, the maximum bias voltage is:

$$v_{b,max} = \frac{2V_{oc}(1-\gamma)}{1+\gamma}, \tag{18}$$

where $\gamma \in (-1, 0)$ is the signed voltage-flipping factor and the corresponding achievable range of electrically induced damping is:

$$\left\{ C_e \in \mathbb{R}^- \;\middle|\; \frac{-4\alpha_e^2\left(3\gamma^4 - 5\gamma^3 + 8\gamma^2 - 7\gamma + 1\right)}{\omega\pi C_p(1+\gamma)^2\left(\gamma^2-\gamma+1\right)} \le C_e < 0 \right\}. \tag{19}$$

Therefore, unlike the external time-varying frequency excitation discussed in the previous section, the BECC realizes a self-powered internal time-varying electrically induced damping regulation mechanism, which provides the second orbit-jump strategy for driving the system from the low-energy orbit to the high-energy orbit.

4.2. Modulation results

As shown in Fig. 2, the hysteresis region (Region 2) contains multiple solutions, namely two stable solutions and one unstable solution at a given frequency. The stability of these solution branches can be determined from the Jacobian matrix of Eq. (9), written as:

$$J = \begin{pmatrix} -\frac{c}{2} & a\left(\frac{(1-k_c)\omega^2-\omega_0^2}{2\omega} - \frac{3k_3a^2}{8\omega}\right) \\ -\frac{1}{a}\left(\frac{(1-k_c)\omega^2-\omega_0^2}{2\omega} - \frac{9k_3a^2}{8\omega}\right) & -\frac{c}{2} \end{pmatrix}, \tag{20}$$

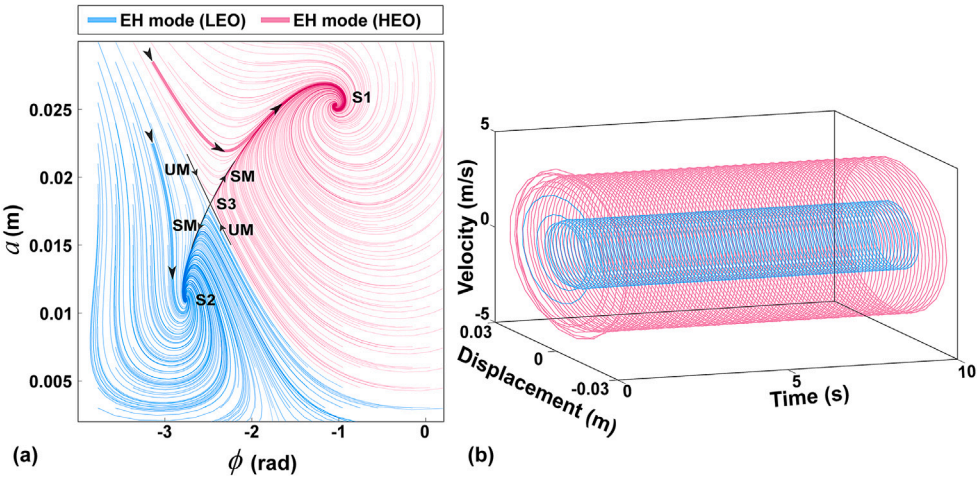


Fig. 8. Phase portrait and phase evolution of the system at a rotational frequency of 9.7 Hz. (a) (a, ϕ) phase portrait; (b) time evolution of the displacement and velocity.

whose eigenvalues are given by

$$\lambda_{1,2} = -\frac{c}{2} \pm \sqrt{\left(\frac{\left(1-k_c\right)\omega^2 - \omega_0^2}{2\omega} - \frac{3k_3a^2}{8\omega}\right)\left(\frac{9k_3a^2}{8\omega} - \frac{\left(1-k_c\right)\omega^2 - \omega_0^2}{2\omega}\right)}. \tag{21}$$

As illustrated in Fig. 8(a), the two stable fixed points, S_1 and S_2, are associated with a pair of complex-conjugate eigenvalues with negative real parts, and therefore exhibit stable spiral behavior. In contrast, the unstable fixed point S_3 possesses two real eigenvalues with opposite signs, indicating a saddle point. Correspondingly, S_3 is associated with a stable manifold (SM) and an unstable manifold (UM). The stable manifold SM divides the (a, ϕ) phase space into two attraction regions, which correspond to the basins of attraction of the two stable fixed points S_1 and S_2. As a result, trajectories starting from different initial conditions on the two sides of SM eventually converge to different attractors, with S_1 corresponding to the HEO and S_2 corresponding to the LEO. This is illustrated in Fig. 8, where two representative initial conditions evolve toward different steady-state orbits.

Based on this phase-space structure, the transition from the LEO to the HEO can be understood as a controlled shift of the system trajectory in the (a, ϕ) plane through internal time-varying electrically induced damping generated by the VE mode of the BECC. Starting from the steady-state LEO, the BECC is first operated in the EH mode, in which the system remains attracted to the LEO fixed point with positive electrically induced damping. Since all trajectories converging to the LEO share the same steady-state amplitude and phase, the corresponding local phase evolution is shown by the light blue trajectories in Fig. 9(a).

The BECC is then switched to the VE mode, which introduces negative electrically induced damping and reduces the overall damping of the system. Under this condition, the system trajectory moves away from the original LEO attractor and evolves toward a new stable fixed point, denoted by S_4, as shown by the yellow trajectory in Fig. 9(a). Because the reduction in damping decreases energy dissipation and promotes response growth, the new stable state S_4 exhibits a larger displacement amplitude than the original HEO fixed point S_1. In other words, the temporary internal damping regulation reshapes the phase portrait and enables the system to cross the original attraction boundary, thereby achieving the orbit jump from the LEO to a high-energy state.

After the HEO is reached, the BECC is switched back to the EH mode, so that the system damping returns to its original level. Consequently, the phase portrait recovers its initial structure, as shown in Fig. 8(a), and the system state is finally attracted to the original HEO fixed point S_1. This return process is represented by the pink trajectory in Fig. 9(a). The corresponding displacement and velocity evolutions are presented in Fig. 9(b), while the temporal variation of the internal damping is shown in Fig. 9(c).

The emergence of the new stable fixed point S_4 in Fig. 9(a) reflects the effect of damping reduction on the local stability structure of the system. As the damping decreases, the real parts of the eigenvalues approach zero, indicating weakened stability and slower convergence of the trajectories. Meanwhile, the reduced dissipation causes the response to evolve toward a higher-energy stable state with spiral asymptotic convergence. In the time domain, this process can be divided into three stages: 0–6 s, during which the system remains on the LEO in EH mode; 6–8 s, during which the VE mode is activated and the orbit jump occurs; and 8–10 s, during which the system returns to EH mode and settles onto the original HEO. These three stages correspond to the differently colored intervals shown in Fig. 9(b) and Fig. 9(c).

5. Experiment

To verify the two orbit-jump strategies proposed in the previous sections, experiments are carried out in this section from both the external-excitation and internal-parameter perspectives. The first objective is to validate whether external time-varying frequency

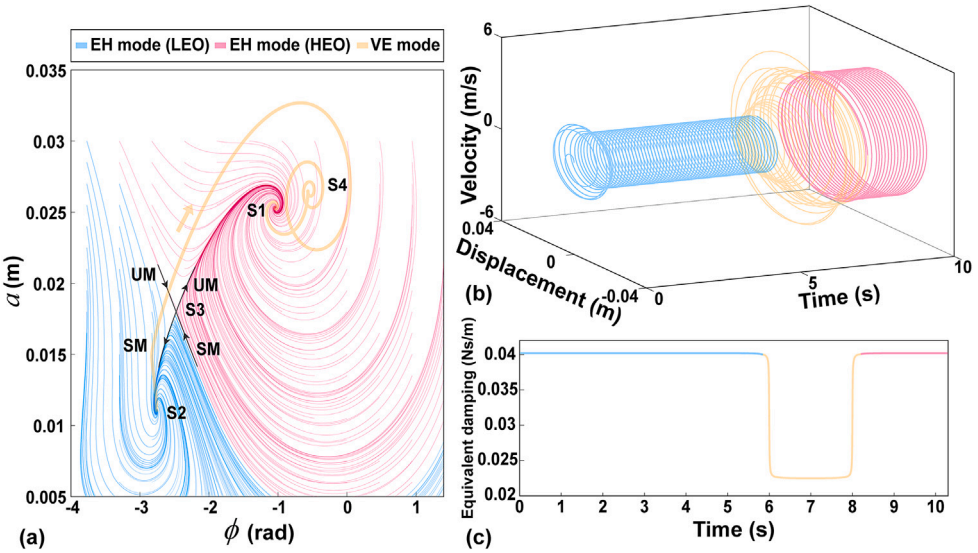


Fig. 9. Phase portrait and phase evolution during an orbit jump at 9.7 Hz. (a) (a, ϕ) phase portrait showing the orbit-jump trajectory; (b) time evolution of the displacement and velocity; (c) time-varying internal system damping.

excitation can reliably drive the rotational harvester from the LEO to the HEO, and to experimentally determine the critical sweep-rate boundary for HEO capture. The second objective is to verify whether the self-powered internal time-varying electrically induced damping, realized by the BECC, can trigger a controlled orbit jump under fixed rotational excitation. Through these two groups of experiments, the present section aims to confirm not only the correctness of the theoretical analyses in Sections 3 and 4 but also the practical feasibility of the proposed orbit-jump framework for rotational nonlinear energy harvesters. Since the negative electrically induced damping in the vibration-excitation mode is supported by the energy harvested and stored in the circuit itself, these experiments further demonstrate the self-powered nature of the internal parameter regulation mechanism.

5.1. Setup

The experimental platform is shown in Fig. 10. It was designed to support the rotational-frequency modulation and electrically induced damping modulation. The nonlinear rotational piezoelectric harvester was mounted on a rotating disk driven by a servo motor (ECM-E2), together with a servo driver (ASDA-B2) and a programmable controller (DKC-Y240). This actuation chain enables prescribed rotational-frequency profiles, including constant-speed operation and time-varying sweep commands, and therefore provides the hardware basis for validating the external time-varying frequency excitation strategy. To monitor the structural response, an IL-030 series CMOS laser displacement sensor was used to measure the transverse displacement of the piezoelectric cantilever beam. The measured analog displacement signal was further processed by a differential circuit to obtain the corresponding velocity signal. This velocity-related signal plays a dual role in the experiments: it is used to characterize the dynamic response of the harvester, and it also serves as the synchronization signal for bias-flip switching in the BECC, since the equivalent piezoelectric current is proportional to the relative vibration velocity. Based on this synchronization signal, a microcontroller (MSP430G2553, Texas Instruments) and a MOSFET switching network were employed to control the BECC operation between the EH mode and VE mode, thereby realizing the internal time-varying electrically induced damping required for orbit-jump experiments. In addition, an oscilloscope (MDO32 C022648, Tektronix) was used to record the displacement readings, piezoelectric voltages, and switching behavior of the interface circuit during operation. As shown in Fig. 10, the platform integrates the rotational structure, piezoelectric beam, dipole magnets, CMOS laser sensing module, differential circuit, BECC, electrical slip ring, and the motor-control unit into a single experimental system. This integrated arrangement enables direct experimental comparison between the two proposed HEO-excitation strategies: one achieved by externally imposed time-varying rotational frequency, and the other by self-powered internal electrically induced damping modulation. The main system parameters used in the experiments are listed in Table 1.

5.2. Experiments on time-varying external frequency modulation

This subsection experimentally validates the first orbit-jump strategy, namely external time-varying frequency excitation. The objective is twofold: first, to verify that a properly designed rotational-frequency sweep can drive the system from the LEO to the HEO; second, to experimentally determine the critical sweep rate that separates successful HEO capture from final convergence to the LEO.

Using the experimental platform shown in Fig. 10, the rotational harvester was driven by the servo motor under programmed frequency-sweep commands. Following the same procedure as in the theoretical analysis, the system was first operated at 8.5 Hz

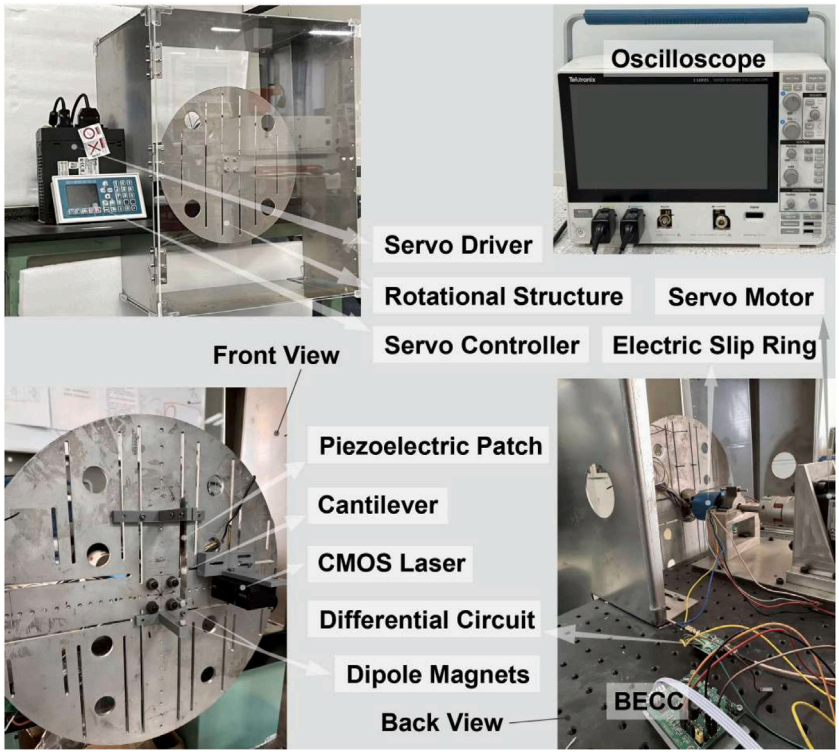


Fig. 10. Experimental setup.

Table 1
System parameters.

Parameter	Value	Parameter	Value
M (g)	3.23	α_e (mN/V)	0.127
C (N s/m)	0.039	κg (mN)	72
K (N/m)	46.21	C_p (nF)	35.33
K_1 (N/m)	4.88	L (mH)	1
K_3 (kN/m^3)	1.92	C_b (μF)	100
MOSFET	ZVU(P)4424	C_{store} (mF)	6.8

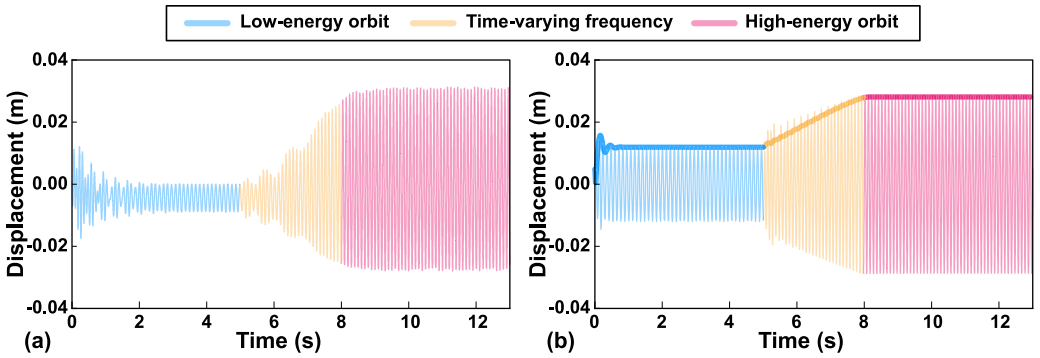


Fig. 11. Comparison of experimental and simulated dynamic responses at f_e = 10 Hz: (a) experimental results; (b) numerical simulation results.

for 5 s to reach a steady initial state on the LEO. The rotational frequency was then increased from 8.5 Hz to the target terminal frequency f_e with a prescribed sweep rate β, after which the system was held at f_e to observe its final orbit. In this set of experiments, the displacement response measured by the CMOS laser sensor was used as the main indicator of whether the system finally settled on the HEO or remained on the LEO.

A representative validation case with f_e = 10 Hz is shown in Fig. 11. When the sweep rate is set to 0.5 Hz/s, the measured displacement amplitude grows continuously during the sweep process and finally stabilizes at a large-amplitude orbit. The experimental response agrees well with the simulated result in both the overall evolution trend and the final response magnitude at the high-energy orbit. This agreement confirms that, under sufficiently slow external time-varying frequency excitation, the system response can continuously follow the high-energy branch and remain locked in the HEO after the sweep is completed. For the classification of the final orbit state, the post-sweep steady-state displacement amplitude was extracted after the rotational frequency reached f_e and was held constant. Specifically, the steady-state amplitude A_{ss} was obtained by averaging the peak displacement amplitude over the final 10 stable vibration cycles of the recorded response. For each terminal frequency f_e, the experimentally identified large-amplitude and small-amplitude steady-state responses were denoted as $A_{HEO}(f_e)$ and $A_{LEO}(f_e)$, respectively. The midpoint between these two amplitudes was used as the classification threshold. If the measured post-sweep amplitude exceeded this threshold, the final response was classified as HEO; otherwise, it was classified as LEO. This procedure was applied to all tested combinations of f_e and sweep rate β.

Although the numerical and experimental responses in Fig. 11 exhibit the same orbit-transition tendency and comparable final HEO amplitudes, visible discrepancies remain during the LEO stage and the transient period associated with the frequency variation. These discrepancies are mainly attributed to small-amplitude nonidealities that are not fully represented in the reduced-order model, including Coulomb friction, nonlinear damping, inaccuracies in the magnetic-force model, and uncertainties in the initial conditions. During the low-amplitude LEO stage, these effects become relatively more significant and may noticeably alter the detailed transient waveform. By contrast, once the response reaches the large-amplitude HEO, the system dynamics are dominated by the nonlinear stiffness and the principal viscous and electromechanical damping terms, resulting in better agreement between the numerical and experimental responses.

To further identify the boundary of successful HEO excitation, a series of frequency-sweep experiments were conducted across the hysteresis region by varying both the terminal frequency f_e and the sweep rate β. For each terminal frequency, the final orbit state was classified using the post-sweep steady-state amplitude criterion described above. The critical sweep rate $\beta_c(f_e)$ was then determined as the largest tested sweep rate at which the final response was still classified as HEO. When the sweep rate exceeded this value, the response could no longer remain on the high-energy branch and finally converged to the LEO. Therefore, the critical sweep-rate boundary shown in Fig. 12 represents the experimentally identified largest successful sweep rate for HEO capture at each terminal frequency. This experimental boundary is consistent with the dynamic-bifurcation interpretation discussed in Section 3.2. When the imposed sweep rate is lower than the local relaxation capability of the nonlinear response, the trajectory can remain in the HEO basin and finally settle on the high-energy branch. In contrast, when the sweep rate is too high, the response lags behind the moving attractor during the frequency-varying process and eventually falls back to the LEO after the sweep is completed. Owing to the discrete sweep-rate increments used in the experiments, the actual transition boundary lies between the largest successful sweep rate and the next failed sweep rate.

It should also be emphasized that the critical sweep-rate boundary is not determined from the detailed transient amplitude during the LEO stage. For each terminal frequency, after the rotational frequency reaches the prescribed terminal value and is subsequently held constant, the final orbit state is classified according to the post-sweep steady-state displacement amplitude. Therefore, this boundary represents the experimentally identified maximum sweep rate at which successful capture into the HEO can still be achieved. The mismatch during the low-amplitude transient stage may affect the detailed time history prior to orbit capture, but it does not alter the final HEO/LEO classification used to construct the critical sweep-rate boundary. Accordingly, the model is employed to elucidate the dominant nonlinear branch structure and the basin-capture mechanism, whereas the quantitative boundary presented in Fig. 12 is determined from the experimentally measured post-sweep steady-state responses.

A typical example is given in Fig. 13 for f_e = 9.9 Hz. When the sweep rate is 5 Hz/s, the post-sweep steady-state amplitude remains on the large-amplitude branch, and the final response is therefore classified as HEO, as shown in Fig. 13(a) and (c). In contrast, when the sweep rate is increased to 5.56 Hz/s, the post-sweep steady-state amplitude falls to the small-amplitude branch, and the final response is classified as LEO, as shown in Fig. 13(b) and (d). Therefore, for f_e = 9.9 Hz, the experimentally identified critical sweep rate is reported as 5 Hz/s, while the actual transition threshold is bounded between 5 Hz/s and 5.56 Hz/s.

Over the entire hysteresis region, the experiments show that HEO excitation can be realized as long as the imposed rotational-frequency sweep remains below the critical sweep-rate boundary. Correspondingly, the proposed external time-varying frequency excitation strategy enables the harvester to operate over an extended high-energy range. As shown previously in Fig. 2, the accessible high-energy operating bandwidth is expanded from 8.8–9.6 Hz to 8.8–10.1 Hz, corresponding to a 62.5% bandwidth enhancement, while the output power is increased by up to 911%.

In summary, the experiments confirm that external time-varying frequency excitation provides a reliable and controllable route for triggering HEOs in the rotational nonlinear energy harvester. More importantly, the experimentally identified critical sweep-rate boundary verifies that the success of HEO capture is governed by the interplay between the imposed frequency-variation rate and the transient response evolution of the nonlinear system. These results provide direct experimental support for the analysis in Section 3 and establish the practical feasibility of frequency-based orbit-jump control.

5.3. Experiments on time-varying internal damping modulation

This subsection experimentally validates the second orbit-jump strategy, namely internal time-varying electrically induced damping realized by the bidirectional energy conversion circuit. In contrast to the external time-varying frequency excitation discussed in the previous subsection, the rotational excitation is kept constant here, and the transition from the LEO to the HEO is

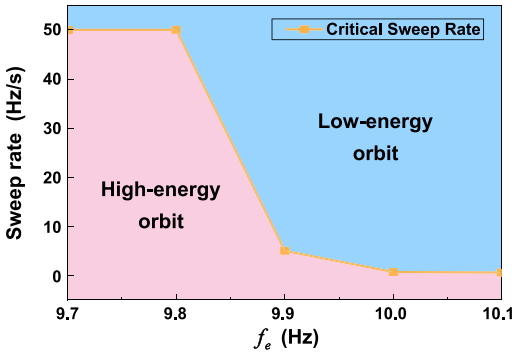


Fig. 12. Experimentally identified critical sweep-rate boundary for HEO capture in the hysteresis region. For each terminal frequency f_e, the critical sweep rate is defined as the largest tested sweep rate at which the post-sweep steady-state response remains on the HEO. Sweep rates above this boundary lead to convergence to the LEO.

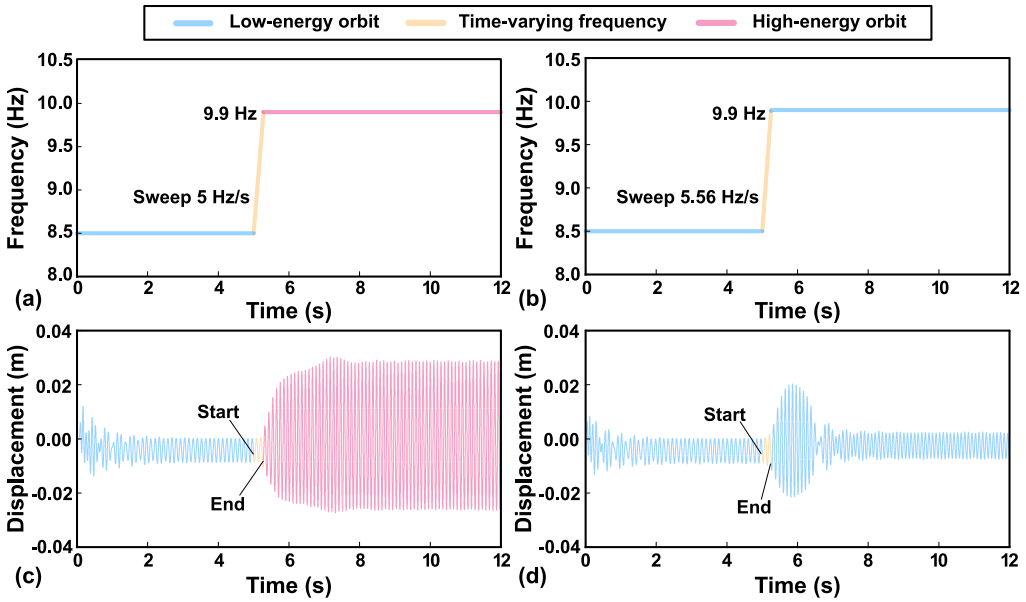


Fig. 13. Experimental comparison of final orbit states at $f_e = 9.9$ Hz. (a, c) Frequency profile and displacement response for $\beta = 5$ Hz/s, where the post-sweep response remains on the HEO. (b, d) Frequency profile and displacement response for $\beta = 5.56$ Hz/s, where the post-sweep response converges to the LEO.

achieved through temporary internal damping regulation. The objective of this experiment is to verify that the BECC can provide a self-powered internal parameter modulation mechanism, allowing the harvester to cross the attraction boundary and finally settle on the HEO under fixed rotational excitation.

A representative orbit-jump experiment under a constant rotational excitation of 9.7 Hz is shown in Fig. 14. Initially, the harvester operates on the LEO with a small vibration amplitude, as indicated by the first light blue segment in Fig. 14. At $t = 1.2$ s, the BECC is switched from the EH mode to the VE mode. In this stage, part of the electrical energy previously harvested and stored in the bias capacitor is fed back to the piezoelectric transducer, thereby introducing negative electrically induced damping and reducing the overall system damping. As a result, the vibration amplitude starts to increase continuously, driving the system trajectory away from the LEO toward the high-energy state. After approximately 7 s of vibration excitation, the response reaches the HEO, as shown in Fig. 14. The BECC is then switched back to the EH mode at $t = 8.2$ s, after which the system remains stable on the HEO while recovering electrical energy at a much higher rate than in the initial LEO state.

The temporal evolution of the piezoelectric voltage during this process is presented in Fig. 15. The measured waveform exhibits the switching behavior predicted by the circuit model under the VE mode, which supports the validity of the proposed electrically induced damping model. As shown in Fig. 14, the effective system damping remains relatively high before 1.2 s, corresponding to the initial EH mode. Once the BECC is switched to the VE mode, the equivalent damping is reduced, and the inset waveform shows that the piezoelectric voltage polarity reverses around the velocity zero-crossing instants. This phase relationship indicates that the

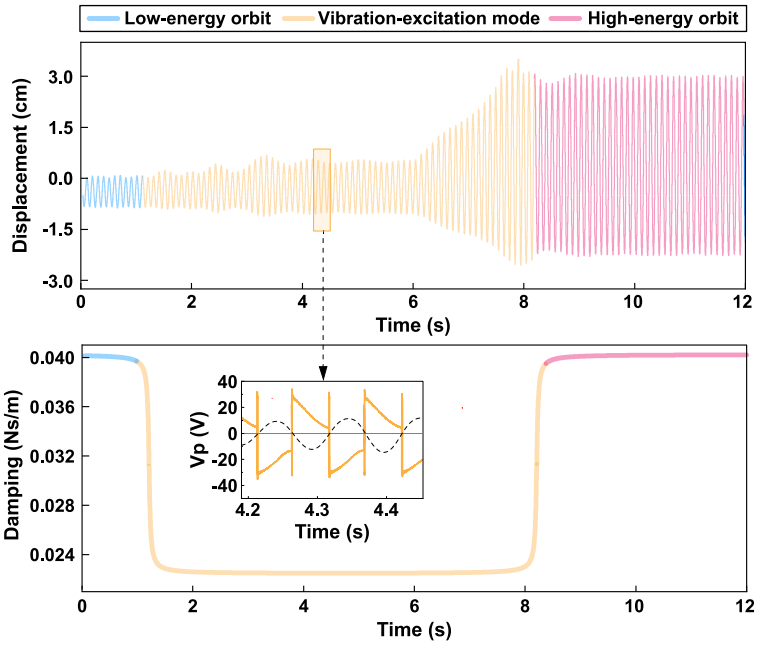


Fig. 14. Experimental demonstration of HEO excitation via time-varying electrically induced damping at a constant rotational frequency of 9.7 Hz. The initial segment corresponds to LEO operation in EH mode, the middle segment corresponds to VE-mode damping reduction and orbit-jump triggering, and the final segment corresponds to stable HEO operation after switching back to EH mode.

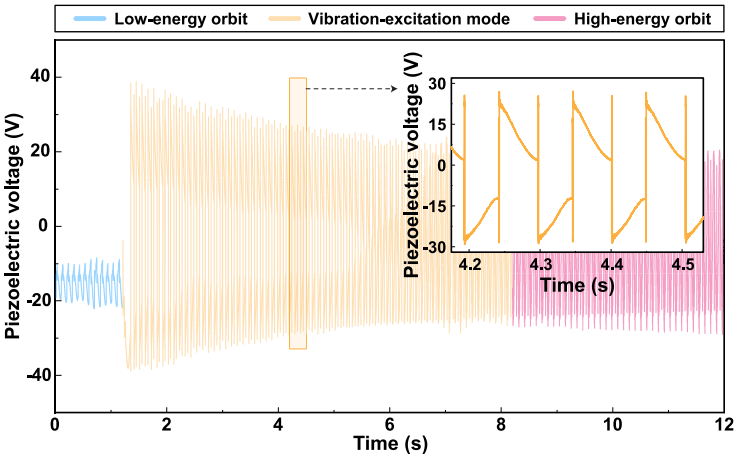


Fig. 15. Time evolution of the piezoelectric voltage in damping-modulation orbit-jump experiments.

stored electrical energy is fed back into the mechanical domain, which promotes response growth and enables the orbit jump. After the HEO is reached, the BECC is switched back to the EH mode, so that the original damping level is restored while the response remains on the HEO.

The energy flow of the BECC-based orbit-jump process can be divided into three stages. In Stage I, the harvester operates on the LEO in the EH mode. The vibration energy is converted into electrical energy by the piezoelectric transducer, and part of the harvested energy is stored in the bias capacitor C_b. In Stage II, the BECC is switched to the VE mode. The stored energy in C_b is partially fed back to the piezoelectric transducer, which generates negative electrically induced damping and drives the response from the LEO toward the HEO. The energy consumed during this VE stage is estimated from the voltage decrease of the bias capacitor as

$$E_{VE} = \frac{1}{2} C_b \left(V_{bf}^2 - V_{af}^2 \right) \approx 16.5 \text{ mJ}, \tag{22}$$

where V_{bf} and V_{af} are the voltages across C_b before and after the VE process, respectively. This energy is not supplied externally, but is taken from the previously harvested energy stored in C_b.

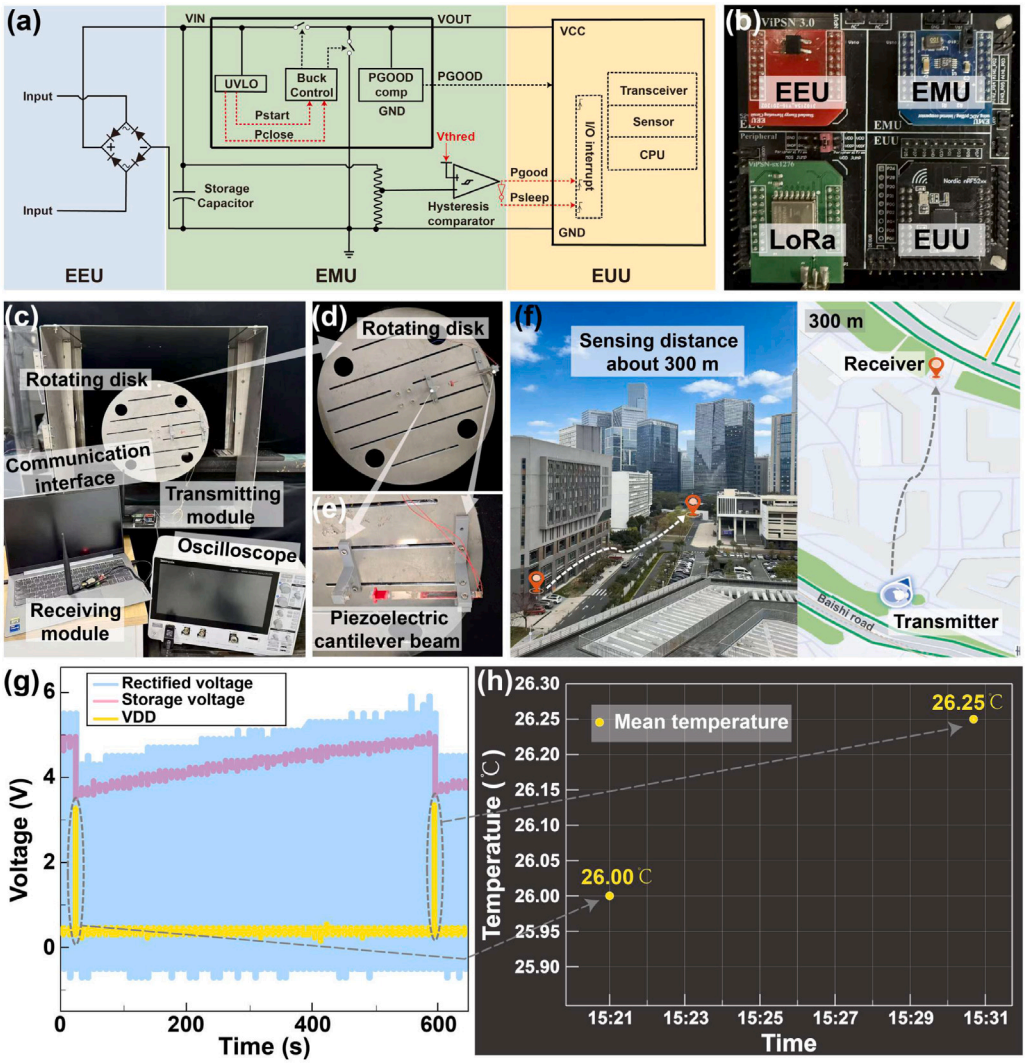


Fig. 16. Application test: (a) topology of the IoT sensor platform; (b) circuit board of the IoT platform; (c)–(e) experimental setup and prototype; (f) sensing scenario and transmission distance; (g) voltage time history; (h) receiving-terminal interface.

In Stage III, after the response reaches the HEO, the BECC is switched back to the EH mode. The system remains on the HEO, and the bias capacitor is recharged by the enhanced piezoelectric output. In the experiment, the harvested power before the orbit jump is approximately 0.05 mW on the LEO. After the system reaches the HEO and returns to the EH mode, the harvested power increases to approximately 0.5 mW. Therefore, the energy consumed during the VE stage can be recovered in approximately $t_{\mathrm{rec}} = E_{\mathrm{VE}}/P_{\mathrm{HEO}} \approx 33$ s. This confirms that the temporary energy consumption in the VE mode can be compensated by the enhanced HEO output over a short recovery period.

5.4. Demonstration of powering a LoRa-based IoT sensing system with the HEO harvester

To further demonstrate the practical significance of the proposed HEO excitation strategies, the rotational nonlinear piezoelectric energy harvester was integrated into a self-powered wireless sensing system for temperature monitoring, as shown in Fig. 16(a) and (b). The demonstration platform is based on the vibration-powered IoT sensor system reported in Li et al. [43] and consists of four main parts: an energy enhancement unit (EEU), an energy management unit (EMU), an energy utilization unit (EUU), and a LoRa communication unit with a temperature sensor. In this configuration, the energy harvester serves as the power source for the entire sensing system.

The alternating electrical output of the harvester is first rectified by a full-wave bridge in the EEU. The rectified energy is then processed by the EMU, which regulates the voltage and stores the harvested energy in a capacitor to intermittently power the sensing and communication electronics. The EMU includes a buck converter, a storage capacitor, and a hysteretic control circuit,

and uses the internal control signals P_{start} and P_{close}, together with the external state signals P_{good} and P_{sleep}, to manage the energy flow and the operating states of the system. Once sufficient energy has been accumulated, the EUU is activated to perform sensing and LoRa-based wireless transmission. When the stored energy drops below the required level, the system returns to the low-power state and waits for the next charging cycle.

The application test was carried out by mounting the rotational harvester on the laboratory rotating platform, while the receiving terminal was placed approximately 300 m away in an outdoor environment, as shown in Fig. 16(f). Because several buildings were located between the laboratory and the receiving terminal, the wireless link involved wall penetration, diffraction, and multipath attenuation. Therefore, this experiment provides a realistic validation of short- to medium-range self-powered wireless communication under non-ideal transmission conditions.

The experiments show a clear difference between the LEO and HEO operating states. When the harvester remains on the LEO, the electrical output is too weak to effectively charge the storage capacitor, and the sensing system cannot be activated. In contrast, after the harvester is driven to the HEO using the proposed orbit-jump strategy, the output voltage and harvested power are significantly enhanced, enabling continuous charging of the storage capacitor and periodic activation of the sensing-transmission module. This result directly demonstrates the value of HEO excitation for practical self-powered applications.

The measured voltage waveforms are shown in Fig. 16(g). The blue curve denotes the rectified harvester voltage, and the pink curve denotes the voltage across the storage capacitor, which fluctuates approximately between 4 V and 5.5 V. After reaching the activation threshold of 5 V, the power supply to the EUU is regulated and maintained as $V_{DD} = 3.3$ V. Once the storage voltage reaches the threshold, the power management circuit enables the buck converter, and the stored energy is released to power one complete sensing-and-transmission event.

The energy consumed during one sensing-transmission event can be estimated from the capacitor discharge process using the same capacitor-energy relation. Here, $C_{store} = 6.8$ mF, $V_{bf} = 5$ V, and $V_{af} = 3.3$ V, which corresponds to an energy consumption of approximately 48 mJ. The active operation takes about 1 s, while the entire charging-execution cycle is approximately 580 s. For the two observed discharge events shown in Fig. 16(g), the LoRa receiving terminal successfully recorded two temperature measurements, namely 26 °C and 26.25 °C, as shown in Fig. 16(h). These results confirm that the HEO-state harvester can provide sufficient energy for intermittent wireless sensing and data transmission, thereby demonstrating the application potential of the proposed orbit-jump-controlled nonlinear rotational energy harvester.

6. Conclusion

This work investigated how to trigger and maintain the high-energy orbit (HEO) in a nonlinear rotational piezoelectric energy harvester through time-varying external and internal regulation. Two complementary orbit-jump strategies were proposed and validated: external time-varying frequency excitation and internal time-varying electrically induced damping.

For the external strategy, the results showed that successful HEO capture is strongly governed by the sweep rate, and that a critical sweep-rate boundary exists in the hysteresis region. When the imposed rotational-frequency variation is sufficiently slow, the system response can remain within the basin of attraction of the HEO and finally stabilize on the HEO branch. For the internal strategy, the bidirectional energy conversion circuit (BECC) was introduced to realize self-powered internal time-varying electrically induced damping. By temporarily generating negative electrically induced damping, the BECC reduces the overall system damping and enables the response to jump from the low-energy orbit (LEO) to the HEO.

Both strategies were confirmed experimentally. The frequency-sweep experiments identified the critical sweep-rate boundary across the hysteresis region, while the BECC-based experiments demonstrated controlled HEO excitation at a fixed rotational frequency of 9.7 Hz. Furthermore, the HEO-state harvester was successfully integrated into a LoRa-based wireless sensing system, providing sufficient energy for periodic temperature measurements and wireless data transmission.

Overall, the results demonstrate that time-varying regulation provides an effective and practical framework for controlled orbit jumps in nonlinear rotational energy harvesters. The proposed external and internal strategies not only deepen understanding of orbit-transition mechanisms in nonlinear systems, but also offer a feasible route for improving the output performance and application potential of self-powered energy harvesting devices.

CRediT authorship contribution statement

Shitong Fang: Writing – review & editing, Writing – original draft, Validation, Supervision, Project administration, Investigation, Funding acquisition, Formal analysis. **Jiabao Zou:** Investigation, Data curation. **Kui Wu:** Writing – original draft, Methodology, Investigation, Formal analysis. **Liuchao Jin:** Writing – review & editing, Visualization, Validation, Data curation. **Mingjing Cai:** Writing – review & editing, Supervision, Investigation. **Xukun Su:** Writing – review & editing. **Xin Li:** Resources. **Zhihui Lai:** Writing – review & editing, Supervision, Project administration, Investigation, Funding acquisition. **Bao Zhao:** Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Software, Resources, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Wei-Hsin Liao:** Writing – review & editing, Supervision, Project administration.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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